

Prateek Arora

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EDUCATION

New York University, NY

December 2024

PhD Candidate, Civil and Urban Engineering, GPA:4/4

Minor in Computer Engineering

Academic Advisor: - Professor Luis Ceferino

Research focused on hazard analysis and urban utilities management in scenarios of extreme events such as hurricanes and earthquakes

Coursework:- Machine Learning, Image and Video Processing, Geomatic Information Systems, Advanced Spatial Analytics, Deep Learning, System Optimization Methods, Energy

Stanford University, CA

March 2020

MS in Structural Engineering, GPA:3.99/4

BITS Pilani, Pilani Campus, India

June 2018

BE(Hons.) in Civil Engineering, GPA:9.54/10.0

Ranked 1st in Civil Engineering class of 90 students, Merit Scholarship

ACADEMIC PROJECTS

Hazard Analysis and Urban Utilities Management under Extreme Events

January 2021 - December 2024

Ph.D. Project

New York University, NY

- Quantifying the impact of power outages on different parts of infrastructure and society
- Predicting the power outages due to high winds of hurricane using machine learning methods

Storm Surge Predictions using Neural Networks

2022 - March 2024

Research in Collaboration with UT Austin

New York University, NY

- Understand the importance of hurricane parameters and bathymetry on storm surge predictions
- Develop neural network based storm surge prediction model to significantly reduce computation time compared to high fidelity models such as ADCIRC

Flood Risk Assessment of New York City

January 2022 - March 2024

Advanced Spatial Analytics

New York University, NY

- Understand the local variability of risk to floods in New York City using Geographic Information Systems
- Utilized Geographically Weighed Regression to map the relation between socio-economic variables and flood risk

Importance Awareness Semantic Segmentation

September 2021 - December 2021

Deep Learning

New York University, NY

- Improved semantic segmentation of urban road scenarios using, SegNet, deep convolutional neural network
- Deployed importance awareness loss to emphasize the safety of pedestrians, and other vehicles, during driving.

Predicting Seismic Response of a Building from Sparse Measurements

January 2020 - March 2020

Random Vibrations

Stanford, CA

- Inferring the dynamic response of a reinforced concrete frame using a sparse array of strong-motion sensing instruments
- Predicting the acceleration time-history response at an uninstrumented floor from the time history of a floor instrumented with accelerometer, using the auto-correlation, cross-correlation, power spectral density, and coherency functions

Graphical User Interface to compute measures of seismic performance

September 2019 - December 2019

Performance Based Earthquake Engineering

Stanford, CA

- Developed Graphical User Interface using MATLAB environment for computation of Mean Annual Frequency of Collapse, Probability of Demolition, Expected Annual Losses, Deaggregation Curve of Collapse Probability and Expected Annual Losses based on Performance Based Earthquake Engineering(PBEE) Framework

HONORS AND AWARDS

Coalition of Disaster Resilient Infrastructure Fellowship, \$15,000 <i>Resilience of Power Grid with Renewables to Hurricanes in a Changing Climate</i>	2022
New York University Tandon School of Engineering Fellowship <i>Doctoral Studies in the field of Disaster Risk and Resilience of Urban Environments</i>	2021
New York University Urban Doctoral Fellowship, \$3000 <i>Fellowship to 12 Doctoral Students working towards the development of Urban Cities</i>	2021
K.C. Mahindra Fellowship, \$3000 <i>Fellowship to 82 out of 1200 applicants from India to pursue graduate studies in USA</i>	2018

EXPERIENCE

Rivera Consulting Group, Inc. <i>Staff Engineer</i>	May 2020 - December 2020 <i>San Francisco, CA</i>
· Analysis and Retrofit of existing Steel-Frame, Unreinforced Masonry, and Wood-Frame Buildings · Designed Connections for different Mechanical Equipment	
Thornton Tomasetti, Inc. <i>Structural Engineering Intern</i>	June 2019 - September 2019 <i>San Francisco, CA</i>
· Checked capacity of Steel Beams, and Connections for Shear Tabs, Non-Structural Components, and Roof Screen. · Reviewed structural steel shop drawings, and rebar shop drawings submittals	
CSIR-Structural Engineering Research Centre <i>Research Intern, Structural Health Monitoring Laboratory</i>	July 2017 - December 2017 <i>Chennai, India</i>
· Applied method for Damage Identification using Statistical Methods, based on residuals for time-series data using Auto-Regressive Moving Average Model and Kolmogorov Statistical Distance. · Implemented robust methods to account for variations in SHM data due to different environmental conditions, segregating the outliers in the data from different origins.	
Mormugao Port Trust <i>Summer Intern, Civil Engineering Department</i>	May 2016 - July 2016 <i>Goa, India</i>
· Performed Costing and Estimation for a Railway Signaling Building at Mormugao Port. · Prepared the report of construction which included placement of formwork, detailing of steel reinforcement.	

ACHIEVEMENTS

- Recipient of O.P. Jindal Engineering and Management Scholarship. awarded for academic and leadership excellence, 2016
- Recipient of Merit Scholarship (awarded to top 3 percent students in a class across all disciplines) at BITS Pilani
- Qualified National Standard Examinations in Chemistry and was among the 300 students from India to be eligible for Indian National Chemistry Olympiad (INChO), 2013
- Secured position among top 1 percent students from Rajasthan State in National Standard Examinations in Physics, conducted by Indian Association of Physics Teacher, 2013

PUBLICATION

Singh M, Yadav S, Hussain S, Arora P, Srivastava A. Reliability-based mix design of marble dust incorporated concrete and its assessment using the concept of performance index. *Structural Concrete*. 2022 <https://doi.org/10.1002/suco.202100501>

Athira G, Sanket R, Aniruddha T, Prateek A, and Bahurudeen A, "Assessment of Sustainable Alternative Cementitious Materials Using Characterisation Techniques", *Proceedings of Civil Engineering Conference-Innovation for Sustainability; CEC- 2016, NIT Hamirpur; 9-10 Sept. 2016; pp 430-445*

TEACHING RESPONSIBILITY

Teaching Assistant, to Professor Luis Ceferino <i>Disaster Risk Analysis and Urban Systems Resilience, CUSP-GX 8006</i>	January 2023 - May 2023 <i>New York, NY</i>
· Led discussion and tutorial sections for class of 13 students · Conducted weekly office hours to assist graduate students with weekly assignments and class project	

Graduate Course Assistant, to Professor Anne Kiremidjian
Introduction to Performance Based Earthquake Engineering, CEE 288

January 2020 - March 2020
Stanford, CA

- Led discussion and tutorial sections for class of 26 students
- Conducted weekly office hours to assist graduate students with weekly assignments and class project

Tutorial Section Leader
Construction Planning and Technology

February 2017 - April 2017
BITS Pilani, India

Delivered Lecture to junior class on designing in Revit Architectural for the Construction Planning and Technology Course

MENTORSHIP

Summer Undergraduate Research Program, New York University

June 2022 - August 2022

- Mentoring two undergraduate students to develop power outage prediction tool and study on scalability of solar energy

Mentor to Master's Student, New York University

January 2022-May 2022

- Mentored graduate students to understand risks to the power system from hurricanes

Seismic Design Competition, Earth Engineering Research Institute

2021-2022

- Mentored undergraduate students for the seismic design competition at National Engineering Earthquake Conference

POSITION OF LEADERSHIP

President, Civil Engineering Association, BITS Pilani

January 2018 - May 2018

Publicity Coordinator, Civil Student Society, BITS Pilani

August 2016 - May 2017

SKILLS

Certification	FE Civil; NCEES Confirmation ID - 20-166-28
Softwares	ArcGIS, MATLAB, Python, ETABS, SAP2000, ABAQUS, ANSYS, MathCAD, RISA, Revit, AutoCAD Machine Learning, MS Project, STAAD.Pro, Adobe Photoshop, Latex, DYNAMO, C++

EXTRA-CURRICULAR ACTIVITIES

- National Level Finalist for Ad-Mania, at Interface (Annual Festival, Department of Management, BITS Pilani) 2017
- Worked for National Service Scheme School and taught students of class 10th and 12th from nearby rural area of Pilani
- Student Member of Indian Concrete Institute, Aug 2016 – Aug 2017
- Volunteered in Blood donation camp and Junoon, a two-day sports meet organized for specially abled people